End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech CSE-AI/ AI&ML , Semester: II
(July, 2023)

Subject Code: BAI-104	Subject: Introduction to Data Science

Time :3 Hours Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	
	(a) Define Data Science Ethics and describe any two ethical issues in Data Science.		
	(b) "Universities use data science to enhance the study experience of their students." Justify the statement using an example.		
	(c) You have created a data frame (by reading an excel file) using pandas named df. You have to replace all missing values in the dataframe with mean of variable.		
	(d) Create an 8X3 integer array from a range between 10 to 34 such that the difference between each element is 1 and then Split the array into four equal-sized sub-arrays.		
	(e) Write a Python program to draw a scatter plot comparing two subject marks of Mathematics and Science. Use marks of 10 students. Sample data: Test Data: math_marks = [88, 92, 80, 89, 100, 80, 60, 100, 80, 34] science_marks = [35, 79, 79, 48, 100, 88, 32, 45, 20, 30] marks_range = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]		
	(f) How can you create a subplot design of a 3x2 design, where the complete first column is spanned?		
	(g) Describe three features of tableau.		
	(h) What is the difference between .twb and .twbx extension?		
	UNIT-I		
Q2	Describe along with examples various types of data in Data Science.	**(10)**	
Q3	i. Explain different ways of combining data from different data sources. Give Examples. ii. What are Dummy Variables? Explain with the help of an example.	**(10)**	
	UNIT-II		
Q4	The super store Dataset includes data for the Sales of multiple products sold by a company along with subsequent information related to geography, Product categories, subcategories, sales, profits, segmentation amongst the consumers, etc. An order is considered a 'high-value' order when both of the following conditions are met: • Sales value of the order is greater than 600 • Profit earned from the order is greater than 90 Write code using Pandas. a. List the all the orders a high-value order with the third highest sales value? b. Which Category is Best Selling and Most Profitable? c. Our DataFrame has a column containing Order IDs in the format IN-2023-Y where Y represents a whole number. An order ID is only valid if Y consists of exactly 9 digits.	**(10)**	

For example, IN-2023-123456789 is a valid Order ID, but IN-2023-123456 (or IN-2023-12345678) is not.
How many invalid Order IDs are there in super store Dataset?

d. Find the number of orders that belong to the category 'furniture' in the top 25 orders based on the sales value.

Q5 (10)

i. Write a NumPy program to create a structured array from given student name, height, class and their data types. Now sort the array on height.

ii. Write a NumPy program to create a 3x3 identity matrix and stack it vertically and horizontally.

iii. Write a NumPy program to create a 3x3 array with random values and subtract the mean of each column from each element.

iv. Write a NumPy program to create a 5x5 array with random values and find the largest value in each row.

UNIT-III

Q6 (10)

Read the csv file sales_data using Pandas :

month_number	facecream	facewash	toothpaste	bathingsoap	shampoo	moisturizer	total_units	total_profit
1	2500	1500	5200	9200	1200	1500	21100	211000
2	2630	1200	5100	6100	2100	1200	18330	183300
3	2140	1340	4550	9550	3550	1340	22470	224700
4	3400	1130	5870	8870	1870	1130	22270	222700
5	3600	1740	4560	7760	1560	1740	20960	209600

i. Read Total profit of all months and show it using a line plot with the following Style properties :
Line Style dotted and Line-color should be red
Show legend at the lower right location.
X label name = Month Number
Y label name = Sold units number
Add a circle marker
Line marker color as read
Line width should be 3

ii. Calculate total sale data for last year for each product and show it using a Pie chart

Q7 (10)

i. Collect data about colleges in Delhi University or any other university of your choice and number of courses they run for Science, Commerce and Humanities, store it in a CSV file and present it using a bar plot.

ii. Collect the minimum and maximum temperature of your city for a month and present it using a histogram plot.

UNIT-IV

Q8 (10)

i. What are the four types of quantitative filters that you can create using Tableau?

ii. What are the five tools in Tableau?

Q9 (10)

What is a Pareto chart?
Use the Sample - Superstore dataset to apply a table calculation to sales data to create a Pareto chart and write steps to answer the following :

Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit

i. Which products (captured by the Sub-Category dimension) account for the most total sales.

ii. Show the percentage of total sales that come from the top products.